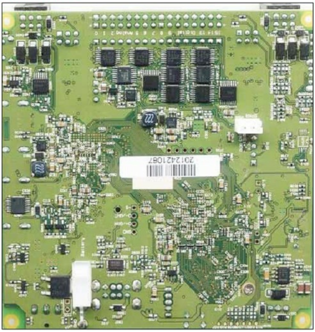
an hour. With each assembly section, there is a Bill Of Materials (listing the parts and tools you will need), general notes for the assembly (what the best working space is, how long it should take, what each part is for, etc), the instruction set itself, and notes on what to do if something goes wrong. Since the project is open source, improvements and features are continually added by builders who have found better solutions, so you should keep checking the website for updates. You are encouraged to contribute to the core system (body, electronics or code) or build extensions (payloads) that plug into the ROV.

A smart robotics controller

http://www.kosagi.com/w/index.php?title=Kovan_Main_Page#Software



Kovan is a smart robotics controller offered by open source hardware company Sutajio Ko-Usagi. It is targeted at applications that require fully autonomous operation and high levels of integration. Kovan integrates onto a single PCB all the features you need to build a self-guided robot. It has high actuator capability supporting four 1.2A H-bridge motor drivers and four servo drivers. It can integrate with a range of sensors via eight 10-bit analog inputs (5V/3.3V selectable range), eight digital I/O (5V/3.3V selectable levels), rapid prototyping headers and a built-in 3-axis accelerometer. Kovan also offers several connectivity options including 802.11b/g Wi-Fi and USB 2.0. Its user interface includes an LCD + touch screen connector (it natively supports an 8.89 cm (3.5") screen), mono audio output, and pushbutton and status LED indicators. It includes an integrated 2-cell Li-ion battery charger.

The open twist: At the heart of Kovan is Linux 2.6.34 running on an 800 MHz Marvell ARM w, 128 MB DDR2 and a 2 GB micro SD for firmware storage. There is a field-programmable gate array (FPGA) co-processor that enables hard real-time control extensions and advanced image-processing extensions. The native development environment for Kovan is based upon OpenEmbedded, the system that is used by BeagleBone. The package comes with C and Python support out-of-the-box, so you can get started right away with development, without any need to set up cross-compilers. The Gerber files, schematics, source code, firmware, etc, are all available on the website; hence, it is a ready-to-use package.

An electronic necklace for the techie woman

http://adafruit.com/inecklace

Although just an accessory, Adafruit's iNecklace is an example of how electronics is entering our life in so many strange yet interesting ways. Made for women who celebrate art, science and technology, the pulsating necklace features a pulsing similar to the 'breathing' LED pattern on many laptop and computer systems. The default pattern is reverse-engineered from Apple's 'breathing' LED on the Mac, MacBook and iMac.

The open twist: The iNecklace is CNC machined from fine aluminium, with a screw-in backing. The pendant contains a circuit board with pulsating LED and battery. A single battery can keep the device pulsing for over 72 hours. The iNecklace hardware is completely open source, so anybody can create their own accessory or gadget using the reverse-engineered pulsing technology. The source code, circuit board files, schematics and CAD files are posted on GitHub (*https://github.com/adafruit/iNecklace*). END

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